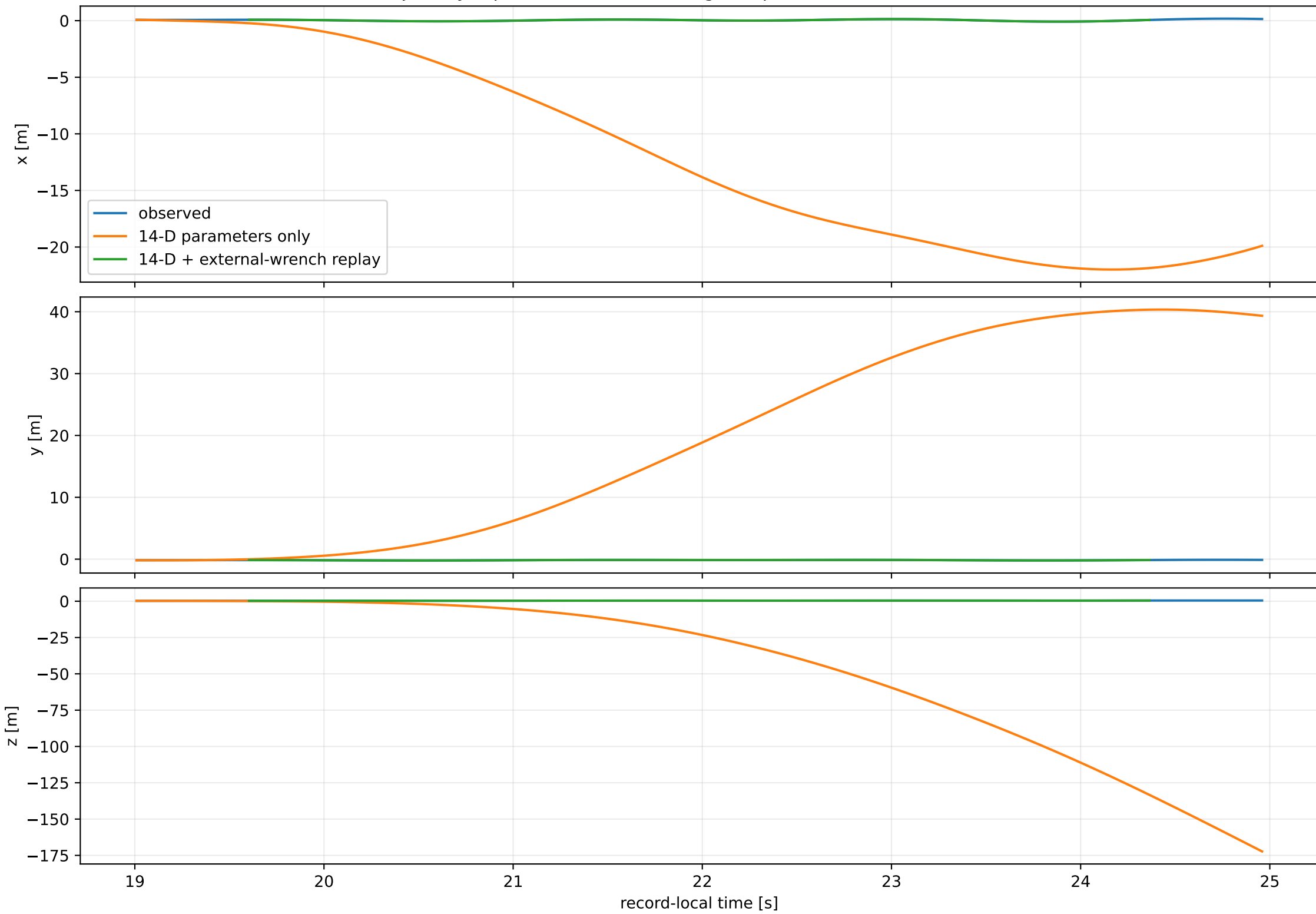
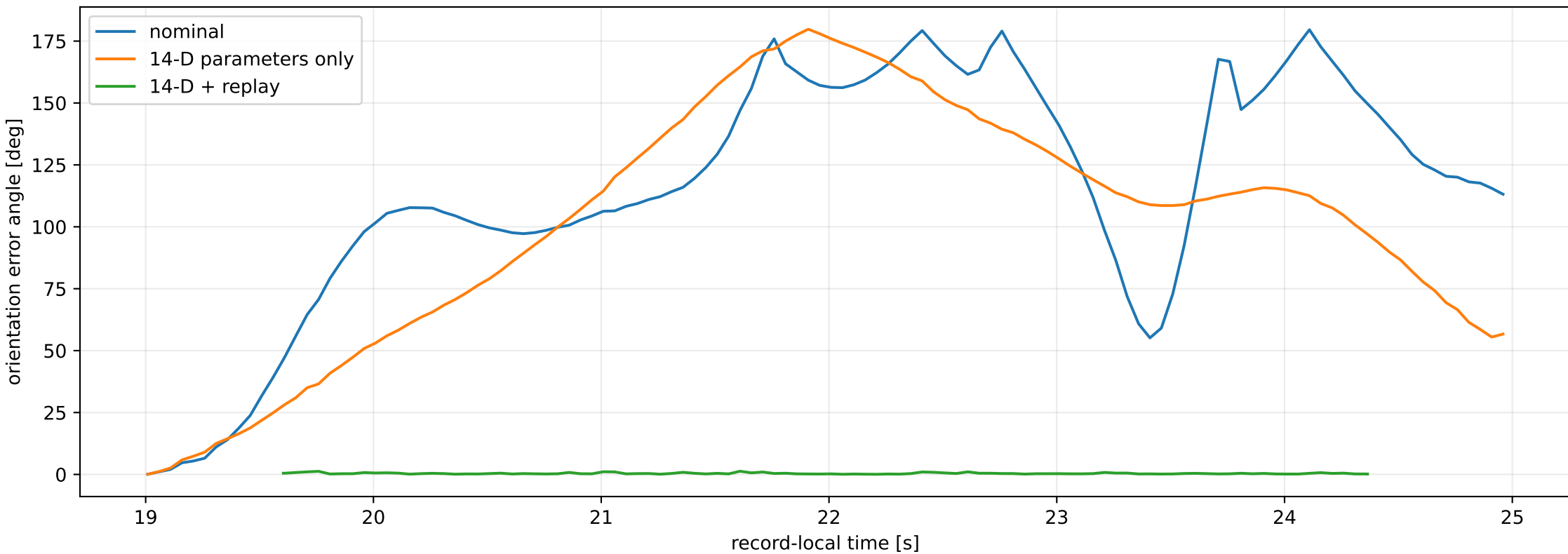
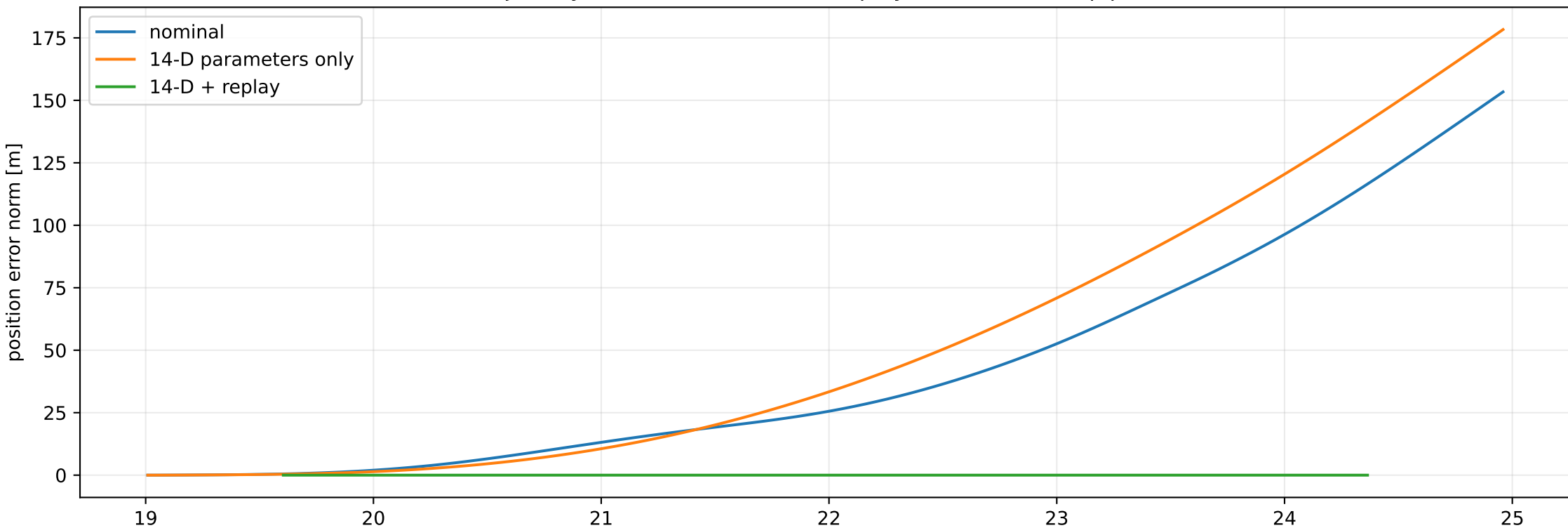


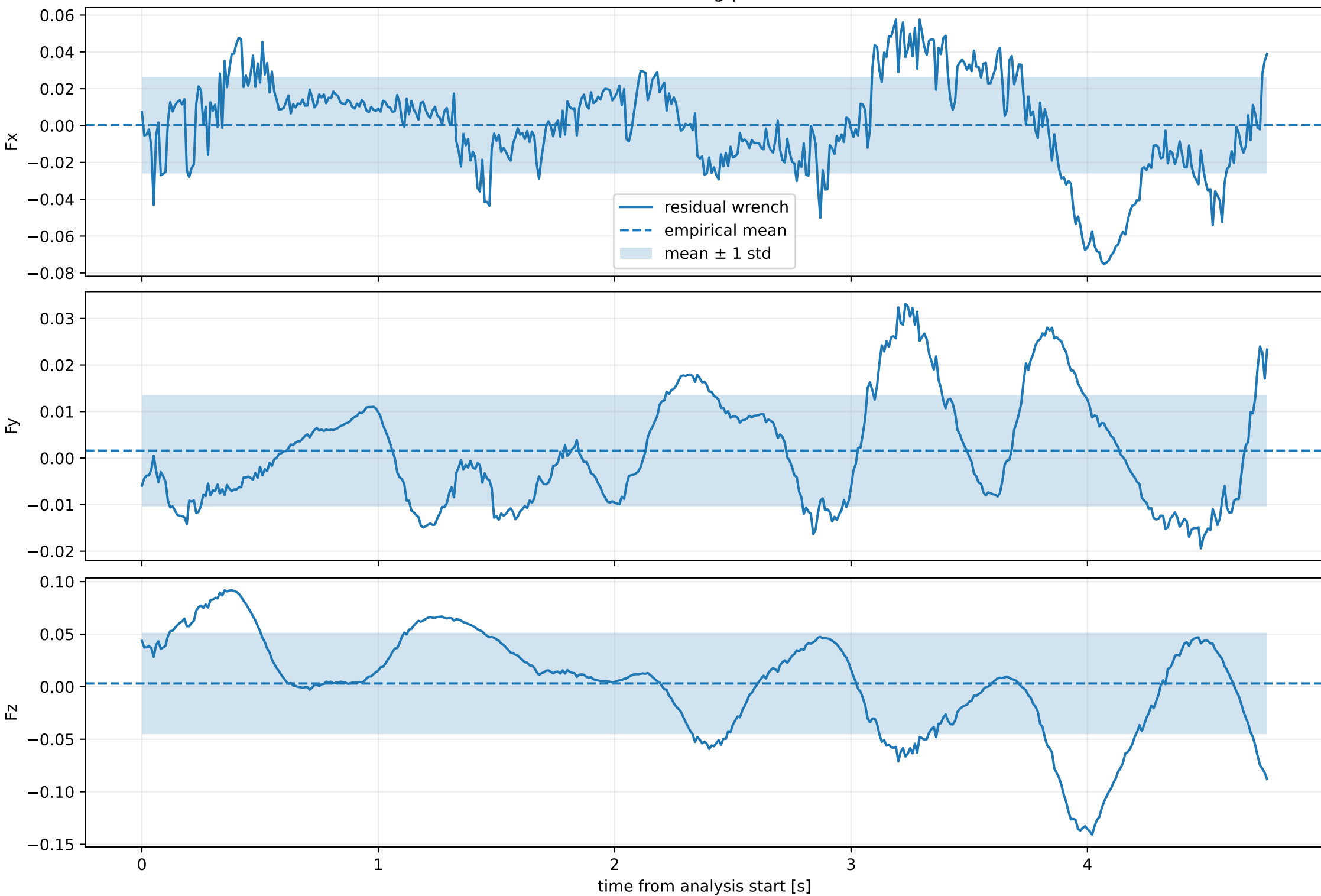
Trajectory reproduction after adding independent rotor effectiveness



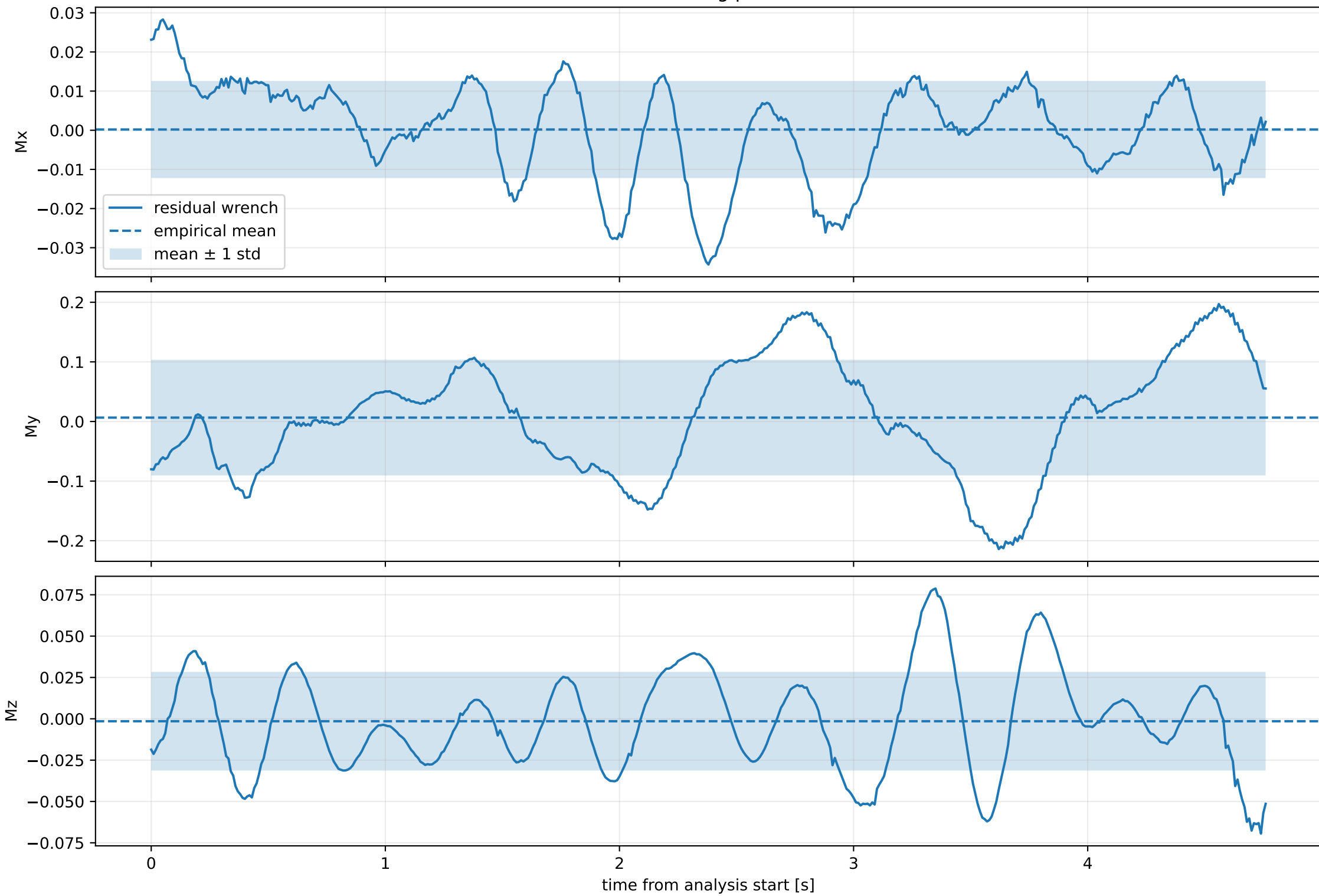
Trajectory error: lower is better; replay verifies the full pipeline



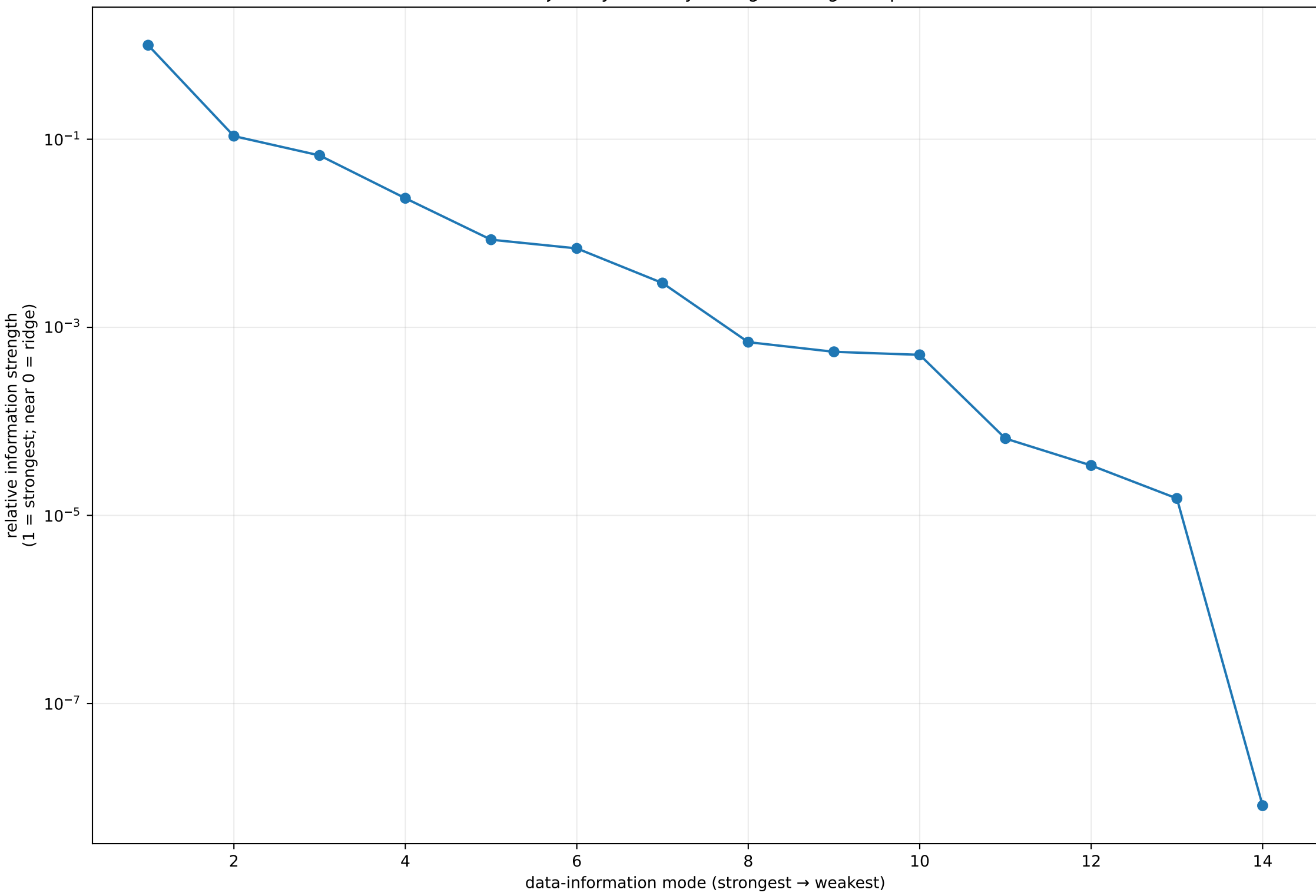
Dimensionless residual force: empirical mean $\pm$ 1 std
Wide band = disturbance varies more, so matching parameter directions receive weaker evidence



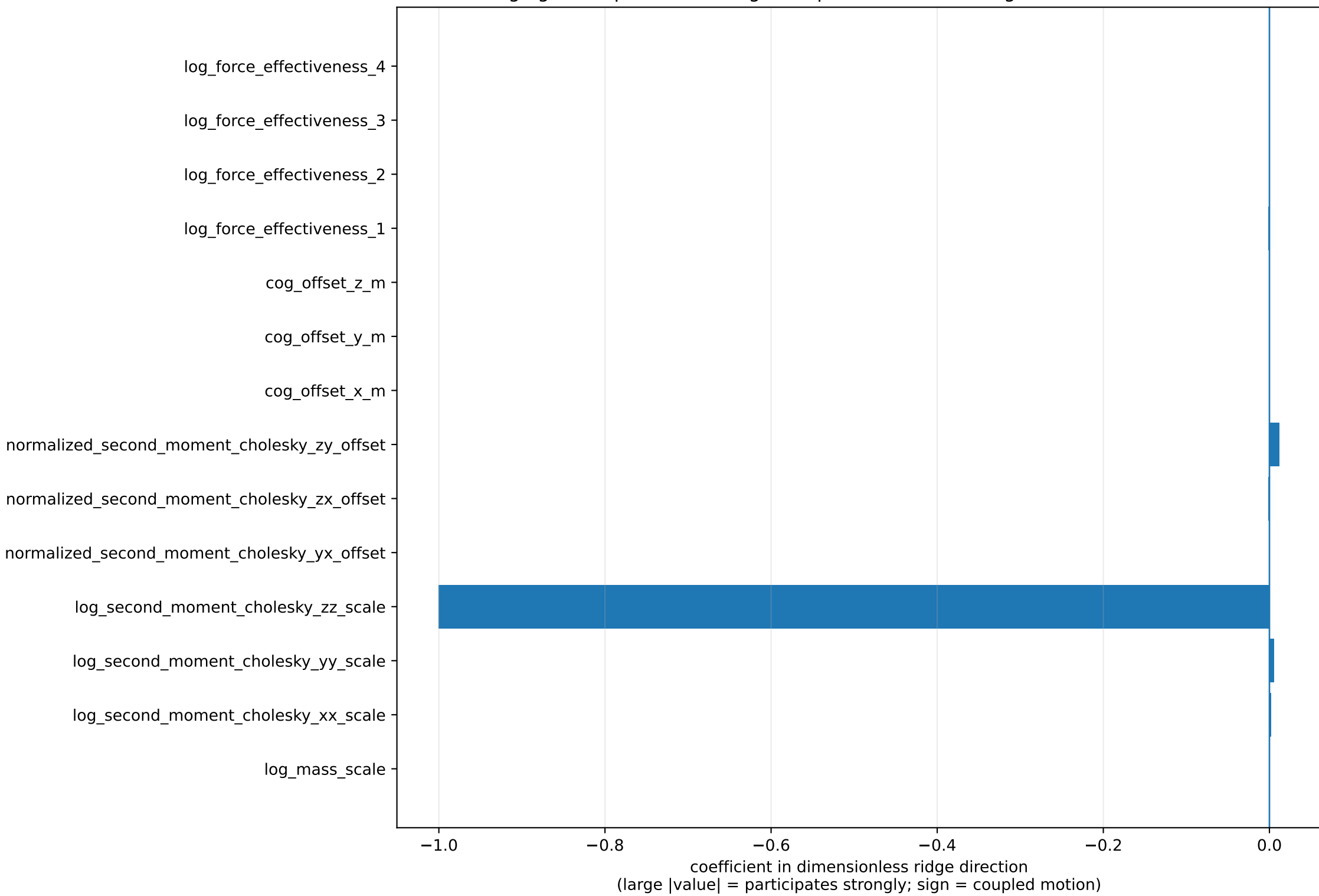
Dimensionless residual torque: empirical mean $\pm$ 1 std
Wide band = disturbance varies more, so matching parameter directions receive weaker evidence



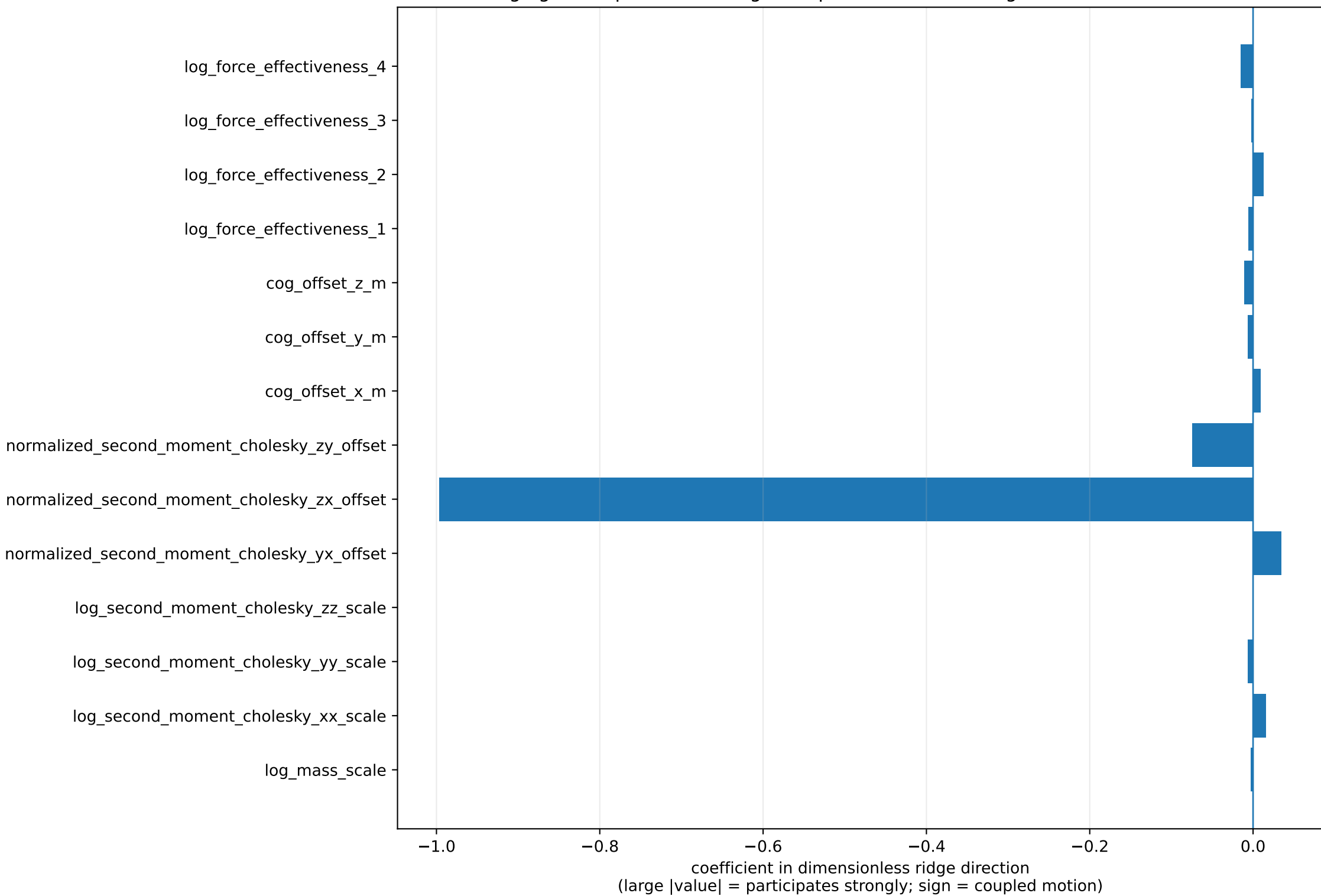
Data-only identifiability spectrum
Small values mean the trajectory scarcely changes along that parameter combination



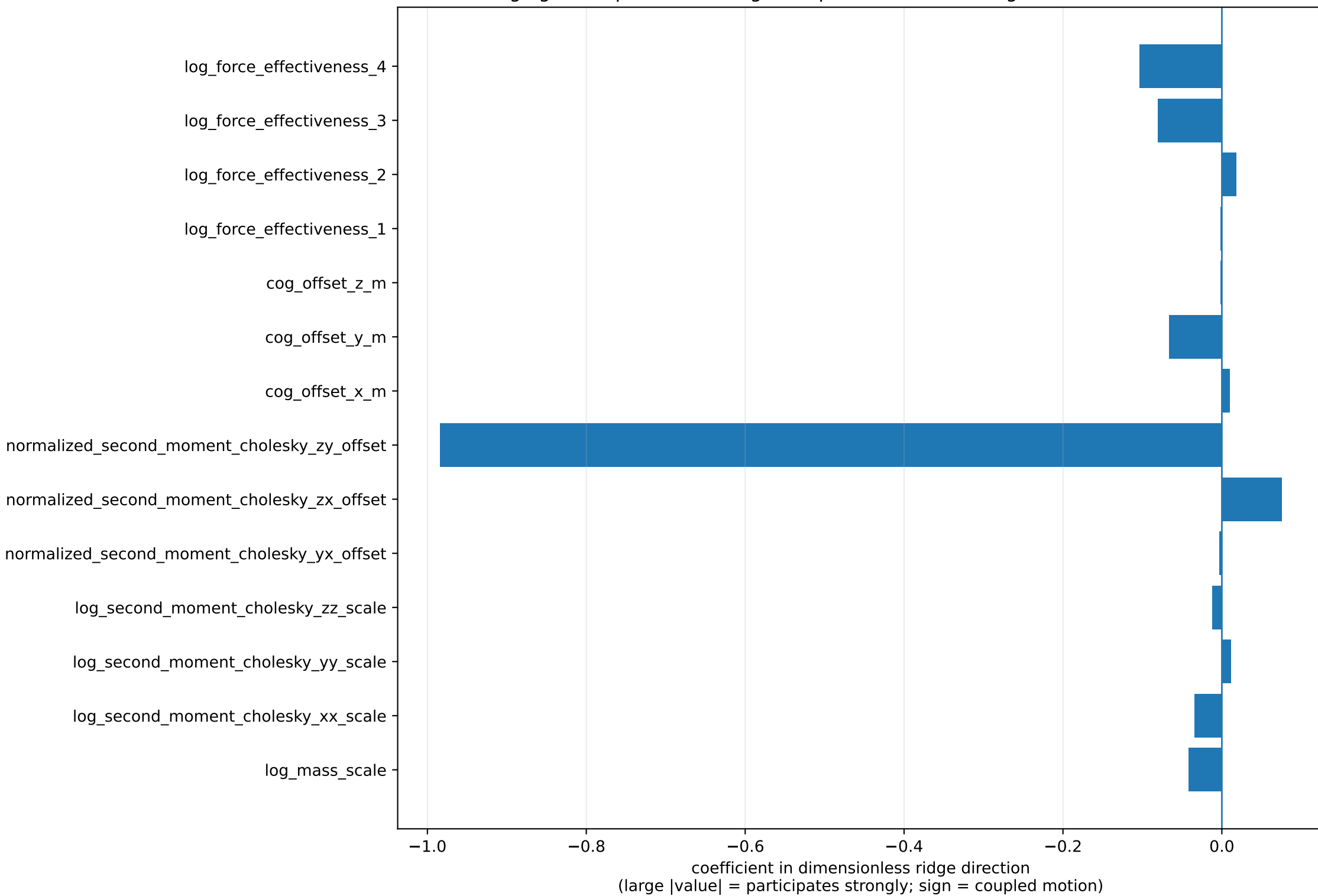
Ridge direction 1: relative information = $8.210\text{e-}09$
Changing these parameters together produces little change in the wrench likelihood



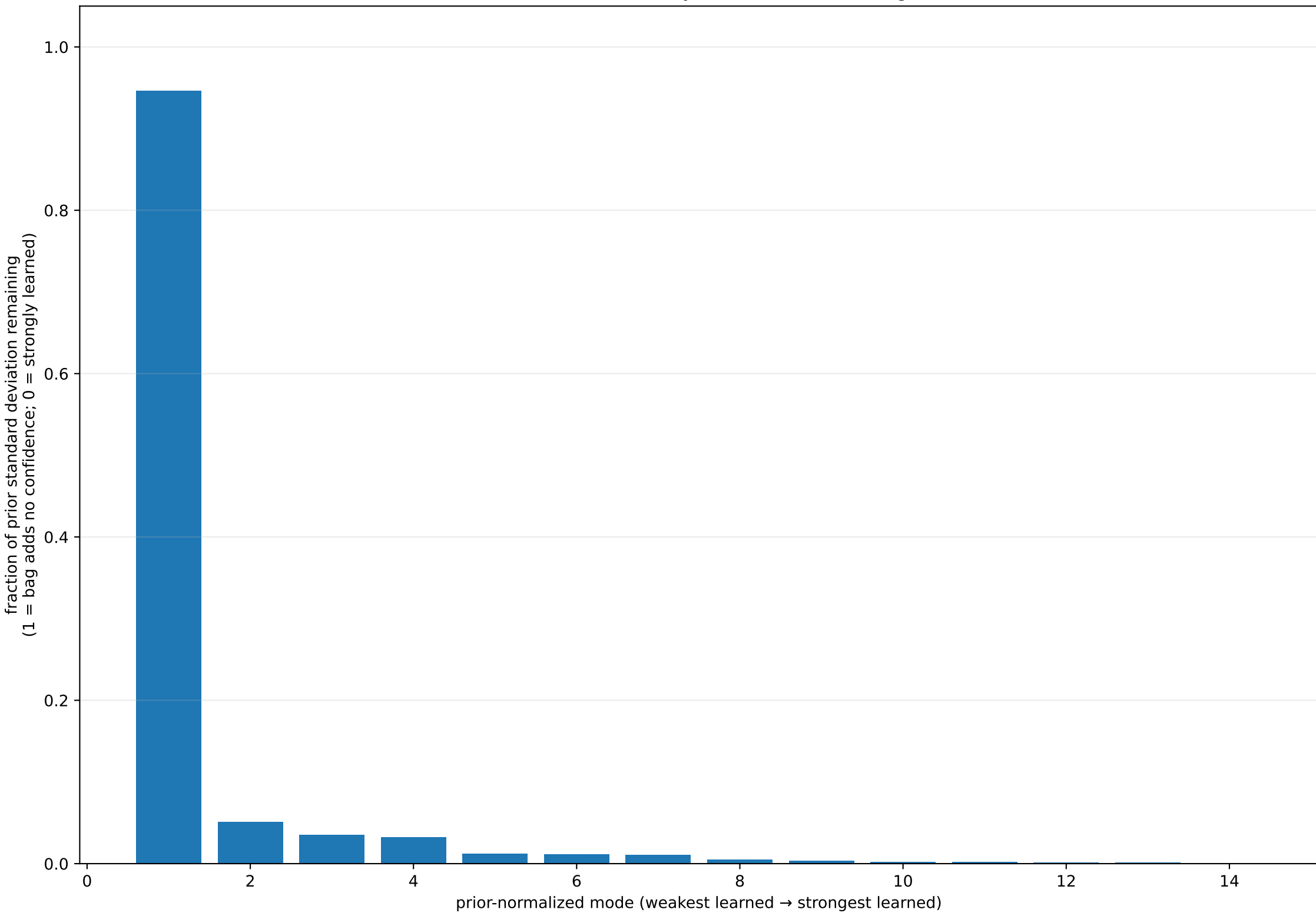
Ridge direction 2: relative information = 1.516×10^{-5}
Changing these parameters together produces little change in the wrench likelihood



Ridge direction 3: relative information = $3.400\text{e-}05$
Changing these parameters together produces little change in the wrench likelihood



How much uncertainty remains after this bag?



Confidence accumulation as the rosbag becomes longer

